Week 7 Material - The Fundamental Theorem of Calculus

28. [1] Evaluate the integral $\displaystyle\int_0^5 \frac{1}{(x+1)^3}\,dx$.

29. [2] Consider the two functions $y_1 = \cos(\pi x)$ and $y_2 = \sin(\pi x)$.

    (a) Write an expression using integrals that defines the total area enclosed between the graphs of y_1 and y_2 over the interval $x \in [0, 1]$

    (b) Evaluate your expression from part (a) to compute the enclosed area.

30. A pump is adding water to a tank at a rate of $r(t) = 20 + 0.1t^2$ liters per minute, over $t = 0 \ldots 10$ minutes. At time $t = 0$ the tank contains 200 liters of water.

    Find the amount of water in the tank after 10 minutes.

31. Evaluate the following integrals, then note any similarities or differences.

    (a) $\displaystyle\int \frac{1}{x^2}\,dx$

    (b) $\displaystyle\int \frac{1}{\sqrt{x}}\,dx$

    (c) $\displaystyle\int \sqrt{x}\,dx$

    (d) $\displaystyle\int \frac{1}{x}\,dx$

32. [3] Evaluate the integral $\displaystyle\int \frac{x^3 - 2}{\sqrt{x}}\,dx$.

33. If $\displaystyle\int_0^9 f(x)\,dx = 37$, and $\displaystyle\int_0^9 g(x)\,dx = 16$, find $\displaystyle\int_0^9 [2f(x) + 3g(x)]\,dx$.

[1] Source: previous APSC 171 Final Exam
[2] Source: previous APSC 171 Test 2
[3] Source: previous APSC 171 Test 2